

Fan Liu

3-237A, Massachusetts Institute of Technology, 77 Massachusetts Ave., Cambridge, MA 02139, USA
Email: liuf@mit.edu • Tel: +1 (617) 347 6996 • Google Scholar • LinkedIn

EDUCATION

Massachusetts Institute of Technology *Cambridge, MA, USA*

- Ph.D. in Mechanical Engineering *Sep 2019 – May 2024*
 - Thesis: Intercellular flow-mediated force relaxation measurement on the three-dimensional multicellular tissue
 - Adviser: Prof. Ming Guo

Tsinghua University *Beijing, China*

- M.S. in Mechanical Engineering *Sep 2016 – Jul 2019*

Beihang University *Beijing, China*

- B.S. in Mechanical Engineering (top 1% graduate) *Sep 2012 – Jul 2016*

RESEARCH APPOINTMENTS & EXPERIENCE

Massachusetts Institute of Technology *Cambridge, MA, USA*

- **Postdoctoral Associate**, Mechanical Engineering *Jan 2025 – Present*
 - Mentors: Prof. Ming Guo and Prof. Roger D. Kamm
 - Developing mechanics–transport assays to quantify brain tissue poroelasticity in aging and Alzheimer’s disease.
 - Investigating how vimentin regulates spheroid poroelasticity and traction-mediated invasion.
- **Doctoral Research**, Mechanical Engineering *Sep 2019 – May 2024*
 - Developed a parallel-plate micromechanics platform and combined experiments with finite-element modeling to establish how intercellular fluid flow governs tissue force relaxation.
 - Developed mechanically driven bulk electroporation for intracellular delivery.

Tsinghua University *Beijing, China*

- **Master’s Research**, Mechanical Engineering *Sep 2016 – Jul 2019*
 - Advisors: Prof. Jia Cheng and Prof. Lihong Ji
 - Developed triboelectric high-voltage platforms for microplasma generation and electro-assisted cell printing.

PUBLICATIONS

Journal Articles

- [13] **Liu F.**, Gao B., Lei L.R., Liu S.N., Li H., and Guo M., “Intercellular flow dominates the poroelasticity of multicellular tissues.” *Nature Physics* 21(8), 1311–1318 (2025). **MIT News**, **MIT Spotlight**
- [12] Du Y., Wang Y.R., Bao Q.Y., Xu X.X., Xu C., Wang S., Liu Q., **Liu F.**, Zeng Y.L., et al., “Personalized Vascularized Tumor Organoid-on-a-Chip for Tumor Metastasis and Therapeutic Targeting Assessment.” *Advanced Materials* 37(6), 2412815 (2025).
- [11] Du Y., Xu X.-X., Yu S.-X., Wang Y.-R., Liu Y., **Liu F.**, Liu W., et al., “Dynamics of endothelial cells migration in nature-mimicking blood vessels.” *Talanta* 277, 126415 (2024).
- [10] **Liu F.**, Yang Z., Yao R., Li H., Cheng J. and Guo M., “Bulk Electroporation for Intracellular Delivery Directly Driven by Mechanical Stimulus.” *ACS Nano* 16(11), 19363–19372 (2022).

- [9] Yang Z.*, Yang Y.*, **Liu F.***, Li B., Li Y., Liu X. et al., “Thousandfold boosting instantaneous current of triboelectric nanogenerator based on decoupled charge pump and discharge tube.” *Nano Energy* 98, 107264 (2022).
- [8] Yang Z., Yang Y., Wang H., **Liu F.**, Lu Y., Ji L., Wang Z.L., and Cheng J., “Charge Pumping for Sliding-mode Triboelectric Nanogenerator with Voltage Stabilization and Boosted Current.” *Advanced Energy Materials* 11(28), 2101147 (2021).
- [7] Yang Z., Yang Y., **Liu F.**, Wang Z., Li Y., Qiu J. et al., “Power backpack for energy harvesting and reduced load impact.” *ACS Nano* 15(2), 2611–2623 (2021).
- [6] Wang Z., Shi Y., **Liu F.**, Wang H., Liu X., Sun R., et al., “Distributed mobile ultraviolet light sources driven by ambient mechanical stimuli.” *Nano Energy* 74, 104910 (2020).
- [5] Huo H.*, **Liu F.***, Luo Y., Gu Q., Liu Y. et al., “Triboelectric nanogenerators for electro-assisted cell printing.” *Nano Energy* 67, 104150 (2020).
- [4] **Liu F.**, Liu Y., Lu Y., Wang Z., Shi Y. et al., “Electrical analysis of triboelectric nanogenerator for high voltage applications exemplified by DBD microplasma.” *Nano Energy* 56, 482–493 (2019).
- [3] Cheng J., Ding W., Zi Y., Lu Y., Ji L., **Liu F.** et al., “Triboelectric microplasma powered by mechanical stimuli.” *Nature Communications* 9, 3733 (2018).
- [2] Huo H., Cheng J., Lu Y., and **Liu F.**, “Characteristics of Voltage Multiplier Circuits Driven by Triboelectric Nanogenerators.” *Mechanical Engineering and Technology* 7(3), 223–233 (2018). (In Chinese.)
- [1] Haris M., Watanabe T., **Fan L.[†]**, Widyanto M.R., and Nobuhara H., “Superresolution for UAV Images via Adaptive Multiple Sparse Representation and Its Application to 3-D Reconstruction.” *IEEE Transactions on Geoscience and Remote Sensing* 55(7), 4047–4058 (2017).

(* Equal contribution; [†]published as Liu Fan.)

Conference Proceedings

- [C1] **Liu F.** and Guo M., “Mechanics of multicellular living systems.” *Proceedings of SPIE 13853, Optical Elastography and Tissue Biomechanics XIII*, 1385304 (2026).

Patents & Patent Applications

- [P2] Li H. and **Liu F.**, “Microscale Biological Force Measurement Platform and Method of Use.” Chinese patent application **CN120668504A** (application no. 202410313368.7). Published September 19, 2025. *Patent pending*.
- [P1] Cheng J., **Liu F.**, Huo H., Zhang Y., Ji L., and Lu Y., “Weight-Reduction Training Rehabilitation Platform.” Chinese invention patent **CN107617190B** (ZL201711049698.6). Granted November 15, 2022.

Manuscript in Submission

- **Liu F.**, Rodriguez C., van Zyl J., et al., Kamm D. R., and Guo M., “Vimentin Regulates Spheroid Poroelasticity to Enable Traction-Mediated Invasion.” *In submission*.

AWARDS & FELLOWSHIPS

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| ▪ Rohsenow Fellowship, Massachusetts Institute of Technology | Jan 2021 |
| ▪ School of Engineering SMA Fellowship, Massachusetts Institute of Technology | Sep 2019 |
| ▪ Outstanding Graduate, Beijing Municipality and Tsinghua University | 2019 |
| ▪ Asian Youth Cultural Ambassador, Tsinghua University | Oct 2018 |
| ▪ National Scholarship for Excellent Academic Performance (2%) | 2013, 2014 |

INVITED TALKS & SELECTED CONFERENCE PRESENTATIONS

Invited Talks

- “Fluid Transport and Mechanics in Living Tissues: From Physical Principles to Disease.”
Invited seminar, School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore (virtual), September 7, 2026.
- “Mechanics of multicellular living systems.”
Invited oral presentation, SPIE Photonics West, Optical Elastography and Tissue Biomechanics XIII, San Francisco, CA, January 17, 2026.
- “Intercellular flow-mediated force relaxation measurement in three-dimensional multicellular tissues.”
Invited seminar, School of Systems Science, Beijing Normal University, Beijing, China, November 14, 2024.

Selected Conference Presentations

- “Vimentin Regulates Spheroid Poroelasticity to Enable Traction-Mediated Invasion.”
Oral presentation (scheduled), Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL, October 21–24, 2026.
- “Intercellular flow dominates the poroelasticity of the multicellular tissues.”
Poster presentation, BMES Annual Meeting, San Diego, CA, October 8–12, 2025.

TEACHING EXPERIENCE

Massachusetts Institute of Technology, Teaching Assistant

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| ▪ Mechanics and Materials I (Undergraduate) | Spring 2024, 2025 |
| ▪ Fluid Mechanics (Graduate) | Fall 2020, 2021, 2023 |
| ▪ Mechanics of Solid Materials (Graduate) | Spring 2023, 2024 |
| ▪ Intermediate Heat and Mass Transfer (Undergraduate) | Fall 2021, 2022, 2023 |
| ▪ Introduction to Finite Element Methods (Undergraduate) | Spring 2021, 2023 |

Tsinghua University, Teaching Assistant

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| ▪ Numerical Analysis (Graduate) | Fall 2017 |
| ▪ Mechanical Design Basis (Undergraduate) | Spring 2018 |

PROFESSIONAL SERVICE

- Journal reviewer: *Applied Physics Letters*, *Biophysics Reviews*, *Materials Letters*, *Bioprinting*, *Biomedical Engineering Advances*, *Nano Energy*, and *Nanotechnology*.

PROFESSIONAL DEVELOPMENT

- **Generative Multiscale Materials Design: Physics, AI, Manufacturing**
MIT Professional Education; short course, 3.0 CEUs. Jun 2026

TECHNICAL SKILLS

- Software: MATLAB, COMSOL, Abaqus, SolidWorks, AutoCAD.
- Experimental: 3D printing, confocal and two-photon microscopy, optical tweezers.

[CV compiled on 2026-09-16]